

ABARNAA SREE N

AI SOFTWARE DEVELOPER

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SUMMARY

AI Software Engineer with hands-on experience building and deploying LLM-powered applications and computer vision systems, from retrieval-augmented pipelines to real-time inference tools. Comfortable across the full stack — Python, Flask, React.js — with a focus on taking AI systems from prototype to working, deployed products.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, SQL

GenAI & LLM Engineering: RAG, Prompt Engineering, LangChain, Semantic Search, FAISS

Applied Computer Vision: OpenCV, MediaPipe, Pose Estimation

Full-Stack Development: React.js, Node.js, Express.js, REST APIs

Databases: MySQL, MongoDB

Developer Tools: Git, GitHub, VS Code, Postman

PROJECTS

Jump Count Detection | [GitHub](#)

- Built a real-time jump-counting system for fitness tracking using computer vision techniques.
- Implemented pose estimation and jump detection with OpenCV and MediaPipe.
- Improved detection accuracy for low-height jumps by 70% through threshold optimization, reducing missed/false counts.

Hostel Management System | [GitHub](#)

- Enhanced a Hostel Management System by digitize attendance tracking and daily hostel operations, replacing manual processes.
- Implemented QR-code and location-based attendance verification for students, reducing attendance-marking errors.
- Added leave management and announcement modules to streamline hostel-warden communication.
- Built using React.js, Node.js, Express.js, and MySQL for a full-stack, scalable solution.

AI Document Summarization System | [GitHub](#)

- Developed an AI-powered document summarization and question-answering system using Retrieval-Augmented Generation.
- Implemented semantic search with FAISS and Sentence Transformers to retrieve relevant document context before generating answers.
- Supported multi-format input (PDF, DOCX, TXT) with AI-generated responses powered by the Gemini API.
- Deployed the full-stack application (Flask, HTML/CSS/JavaScript) on Hugging Face Spaces for public access.

OPEN SOURCE CONTRIBUTOR

- Contributed 30+ PRs (10 merged) across AI/ML and cloud infrastructure repositories including Local Deep Research, AnythingLLM, Floci, OpenHands, and RAGFlow.
- Fixed a LangChain text-splitter import bug in Local Deep Research by enforcing parent-first import order, merged after maintainer review.
- Implemented AWS Cognito MFA preferences and fixed a DynamoDB list-append bug in cloud-native applications.
- Resolved IAM policy and Firehose compression issues in CloudFormation, and raised the result limit for a Bedrock AgentCore integration.

INTERNSHIP

Artificial Intelligence Intern – Pinnacle Labs:

July 2025 - Aug 2025

- Built an AI-powered real-time translator with speech recognition, translation, and text-to-speech capabilities using Python, Flask, and React.js.

EDUCATION

Bachelor of Technology in Information Technology

National Engineering College, Kovilpatti

CGPA: 8.71

Aug 2023 - Mar 2027

Higher Secondary Education

Kamlavati Senior Secondary School (CBSE)

Class X : 88.6% | Class XII : 80%

Jan 2013 - Jan 2023

ACHIEVEMENTS

- Secured First Position in Code Debugging at KPR Institute of Engineering and Technology.
- Secured First Prize in the Code Sprint competition organized by the Computer Society of India at National Engineering College.